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Simple related-work comparison

Aspect	Fixed resistive loads	Linear electronic loads	Digital / MCU-based PELs	Our proposed PEL
Control of load current	Single fixed value for given voltage; no programmability.	Continuous control but mostly analog knobs; still manual.	Fully programmable profiles and modes.	Closed-loop PWM control with NE555 + LM358; user-set current via rotary switch.
Response to voltage change	Current changes directly with voltage; cannot maintain constant current.	Can regulate but with significant dissipation and heating.	Can regulate precisely with software control.	Maintains regulated current using buck-type switching stage and feedback.
Efficiency and power loss	Low cost but poor flexibility; wastes power at only one operating point.	Accurate but inefficient; large heat dissipation in pass elements.	Good but depends on design; hardware and firmware more complex.	High-efficiency switching buck converter with MOSFET + inductor and Schottky diode.
Complexity and cost	Very simple, very cheap, but not versatile.	More components, heat sinking, analog design effort.	Most complex: MCU, firmware, interfaces, protection logic.	Simple analog ICs (NE555, LM358, IRLZ44N) – no firmware, low cost, easy to build.
Protection and indication	Typically no overcurrent protection or status indication.	May have protection, but adds extra circuitry.	Can implement advanced protections in software, needs careful design.	Hardware overcurrent comparator, green/red LEDs for normal/fault, adjustable trip via RV2.
Integration with existing lab kits	Often requires changing resistors for each test; not retrofittable as an active module.	Usually standalone commercial units; not designed to drop into student kits.	Typically standalone instruments with digital UI.	Designed as a retrofittable load module to protect and test existing lab kits without changing their circuits.